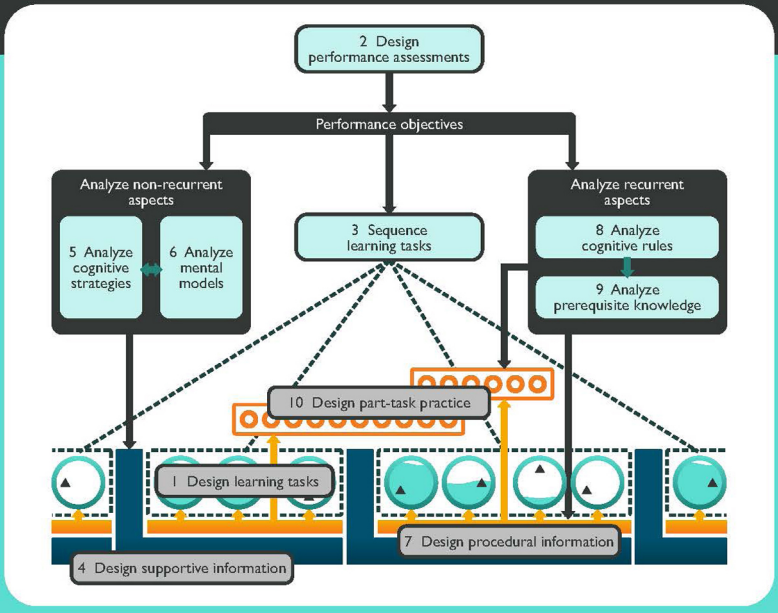
TEN STEPS TO COMPLEX LEARNING

FOURTH EDITION



A SYSTEMATIC APPROACH TO FOUR-COMPONENT INSTRUCTIONAL DESIGN

JEROEN J. G. VAN MERRIËNBOER
PAUL A. KIRSCHNER
JIMMY FRÈREJEAN

Ten Steps to Complex Learning

Ten Steps to Complex Learning presents a path from an educational problem to a solution in a way that students, design practitioners, and researchers can understand and easily use. Students in the fields of instructional design and the learning sciences can use this book to broaden their knowledge of the design of training programs for complex learning. Practitioners can use this book as a reference guide to support their design of courses, curricula, or environments for complex learning.

Driven by the acclaimed Four-Component Instructional Design (4C/ID) model, this fourth edition of *Ten Steps to Complex Learning* is fully revised with the latest research, featuring over 50 new references. The entire book has been updated for clarity, incorporating new, colorful graphics and diagrams, and the guiding example used throughout the book is replaced with a training blueprint for the complex skill of 'producing video content.' The closing chapter explores the future development of the Ten Steps, discussing changes in teacher roles and the influence of artificial intelligence.

Jeroen J. G. van Merriënboer (1959–2023) was Emeritus Professor of Learning and Instruction at Maastricht University, The Netherlands. Before retiring, he was Research Director of the Graduate School of Health Professions Education (SHE) at Maastricht University. He also held honorary positions at the University of Bergen, Norway, and the Open University of the Netherlands. He has published over 450 journal articles and book chapters in the areas of learning and instruction and medical education.

Paul A. Kirschner (1951) is Emeritus Professor of Educational Psychology at the Open University of the Netherlands as well as Honorary Doctor (Doctor Honoris Causa) at the University of Oulu, Finland, and Guest Professor at Thomas More University of Applied Sciences, Belgium. He owns his own educational consultancy company, kirschner-ED, and has published more than 400 journal articles, books, and book chapters in the areas of educational psychology, learning, and instruction.

Jimmy Frèrejean (1986) is Assistant Professor at the Faculty of Health, Medicine and Life Sciences at Maastricht University, The Netherlands. Alongside his teaching and research on instructional design at the School of Health Professions Education, Jimmy actively contributes to the Maastricht University Medical Center and Academy as an educational consultant. He specializes in simulation-based education, coordinates a national lifelong learning program for healthcare professionals, and is part of the TRISIM expert group on training, research, and innovation in simulation-based education in healthcare.



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Ten Steps to Complex Learning

A Systematic Approach to Four-Component Instructional Design

Fourth Edition

Jeroen J. G. van Merriënboer, Paul A. Kirschner, and Jimmy Frèrejean

Routledge
Taylor & Francis Group
NEW YORK AND LONDON

Designed cover image: © Jeroen J. G. van Merriënboer, Paul A. Kirschner, and Jimmy Frèrejean

Fourth edition published 2025
by Routledge
605 Third Avenue, New York, NY 10158

and by Routledge
4 Park Square, Milton Park, Abingdon, Oxon, OX14 4RN

Routledge is an imprint of the Taylor & Francis Group, an informa business

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First edition published by Routledge 2007

Third edition published by Routledge 2017

ISBN: 978-1-032-34508-6 (hbk)
ISBN: 978-1-032-33311-3 (pbk)
ISBN: 978-1-003-32248-1 (ebk)

DOI: 10.4324/9781003322481

Typeset in Galliard
by Apex CoVantage, LLC

To Jeroen, our colleague, mentor, teacher, example, and above all, friend.

Your wisdom and guidance inspired all who knew you, leaving an indelible mark on them and the field.

We will sorely miss you!



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About the Authors

Jeroen J. G. van Merriënboer (1959–2023) was Emeritus Professor of Learning and Instruction at Maastricht University, The Netherlands. Before retiring, he was Research Director of the Graduate School of Health Professions Education (SHE) at Maastricht University. He also held honorary positions at the University of Bergen, Norway, and the Open University of the Netherlands. He obtained a master's degree in experimental psychology from the Vrije Universiteit Amsterdam (1984) and a PhD degree in instructional technology from the University of Twente (1990). Van Merriënboer specialized in cognitive architecture and instruction, instructional design for complex learning, holistic approaches to instructional design, and the use of ICT in education. He has published more than 450 journal articles and book chapters in the areas of learning and instruction and medical education. More than 50 PhD candidates completed their thesis under his supervision. He served on the editorial board of highly ranked scientific journals, such as *Cognitive Processing*; *Computers in Human Behavior*; *Educational Research Review*; *Educational Technology Magazine*; *Educational Technology Research and Development*; *International Journal for Virtual and Personal Learning Environments*; *Journal of Computing in Higher Education*; *Learning and Instruction*; and *Technology Instruction Cognition and Learning*. His prize-winning monograph, *Training Complex Cognitive Skills* (1997), describes his four-component instructional design model for complex skills training and offers a systematic, research-based approach to designing environments for complex learning. He received several awards and prizes for his scientific work; apart from prizes for publications and PhD supervision, he was declared World Leader in Educational Technology by Training Magazine and received the International Contributions Award from the Association for Educational Communications and Technology.

Paul A. Kirschner (1951) is Emeritus Professor of Educational Psychology at the Open University of the Netherlands as well as Honorary Doctor

(Doctor Honoris Causa) at the University of Oulu, Finland, Guest Professor at Thomas More University of Applied Sciences in Belgium, and owns his own educational consultancy company, kirschner-ED. Before retiring, he was University Distinguished Professor at the Open University of the Netherlands and Visiting Professor of Education with a special emphasis on Learning and Interaction in Teacher Education at the University of Oulu, Finland. He has published over 400 scientific articles in the areas of learning and instruction and, at the moment of this writing, has supervised 46 PhD candidates who have successfully completed their theses under his supervision. He is an internationally recognized expert in the fields of educational psychology and instructional design. He was Research Fellow of the Netherlands Institute for Advanced Study in the Humanities and Social Science. He was President of the International Society for the Learning Sciences (ISLS) in 2010-2011 and is Research Fellow of the American Educational Research Association (the first European to receive this honor) and Fellow of the International Society for the Learning Sciences. He was a member of the Scientific Technical Council of the Foundation for University Computing Facilities (SURF WTR) in the Netherlands and was a member of the Dutch Educational Council and, as such, was advisor to the Minister of Education (2000–2004). He is Chief Editor of the *Journal of Computer Assisted Learning* and Commissioning Editor of *Computers in Human Behavior*. In addition to this book, he is also (co)author of a number of very successful books, including *How Learning Happens: Seminal Works in Educational Psychology and What They Mean in Practice* and *How Teaching Happens: Seminal Works in Teaching and Teacher Effectiveness and What They Mean in Practice*, *Evidence Informed Learning Design*, as well as two different volumes of *Urban Legends about Learning and Education*. He also coedited two other books, *Visualizing Argumentation* and *What we know about CSCL*. His areas of expertise include how we learn and what that means for instruction, instructional design, collaboration for learning (computer supported collaborative learning), and the use of media in teaching and learning.

Jimmy Frèrejean (1986) is Assistant Professor at Maastricht University's Faculty of Health, Medicine and Life Sciences, specializing in instructional design within health professions education. He holds a master's degree in work and organizational psychology from Maastricht University and a PhD in instructional design from the Open University of the Netherlands. During his PhD, Jimmy first learned about 4C/ID and the Ten Steps and has continued studying and researching it ever since. He collaborated closely with Professor Jeroen J. G. van Merriënboer in teaching

master's and post-master's courses, conducting research, publishing papers and book chapters, and supervising PhD students at the School of Health Professions Education. Alongside his teaching and research responsibilities, Jimmy actively contributes to the Maastricht University Medical Center and Academy as an educational consultant, helping improve training programs for healthcare professionals. He specializes in simulation-based education, coordinates a national lifelong learning program for healthcare professionals (SimNEXT), and is part of the TRISIM expert group on training, research, and innovation in simulation-based education in healthcare. Jimmy is committed to carrying forward the insights presented in this book, upholding the legacy that shaped the landscape of instructional design.

Preface

More than 30 years ago, the award-winning article *Training for Reflective Expertise: A Four-Component Instructional Design Model for Complex Cognitive Skills* (van Merriënboer, Jelsma, & Paas, 1992) described the first precursor of the *Ten Steps to Complex Learning*. Five years later, in 1997, Jeroen J. G. van Merriënboer published his award-winning book, *Training Complex Cognitive Skills*. That book presented a comprehensive description of a training design system for acquiring complex skills or professional competencies, based on research conducted on the learning and teaching of knowledge needed for jobs and tasks in a technologically advanced and quickly changing society. The basic claim was that educational programs for complex learning must consist of four basic interrelated components: learning tasks, supportive information, procedural information, and part-task practice. Each component was linked to a fundamental category of learning processes. The instructional methods prescribed for each component were derived from a broad body of empirical research.

Whereas *Training Complex Cognitive Skills* was very well received in the academic field of learning and instruction, practitioners in the field of instructional design frequently complained that they found it difficult to systematically design educational programs based on the four components. The article *Blueprints for Complex Learning: The 4C/ID Model* (van Merriënboer, Clark, & De Croock, 2002) was the first attempt to provide more guidance on this design process. The systematic approach described in this article was further developed in the first edition of the book *Ten Steps to Complex Learning* (2007), which can best be seen as a complement to the psychological foundation described in *Training Complex Cognitive Skills*. The Ten Steps describe a path from a training problem to a training solution in a way that students, practitioners (both instructional designers and teachers), and researchers can understand and use. This book was a great success and has been translated into Korean, Chinese, and, in part, in Spanish and Persian. It also spawned a Dutch language analog, of which a new edition appeared at the time of publication of this book.

Sadly, Jeroen J. G. van Merriënboer got very ill and passed away before completing this fourth edition. Despite this, Jeroen played a pivotal role in overseeing and contributing to all the modifications. We want to clarify that any changes made after his passing were discussed with him beforehand. In short, the updates to this book come in three categories. First, this fourth edition has been completely updated with the newest insights into teaching, training, and instructional design. More than 50 new references have been added, and, where relevant, the latest insights from the field have been included. This can be seen in all the chapters in the book. We have also updated the text about cognitive load theory, especially the concepts of intrinsic and extraneous cognitive load.

Second, we have significantly changed the example training-blueprint that formed the book's backbone. In the first three editions, we used the moderately complex skill of 'searching for literature' by a librarian or documentalist to explain and illustrate most of the Ten Steps. This blueprint is now replaced with a more appealing one for a training program for the complex skill of 'producing video content' by a video content producer. The new blueprint is more extensive and detailed, giving a better example of the Ten Step's design approach.

Finally, many smaller and larger changes and additions were made throughout the book to improve readability and comprehensibility. Also, where the third edition included some occurrences of *(s)he*, this is now replaced with *they*. Concrete examples and cases were added or updated where useful. In addition, several figures and tables have been added or revised, and all figures now appear in color. The final chapter consists of an updated perspective on the future of the Ten Steps, including the role of artificial intelligence in developing instruction and training.

The structure of this book is straightforward. Chapters 1, 2, and 3 concisely introduce the *Ten Steps to Complex Learning*. Chapter 1 presents a holistic approach to the design of instruction for achieving the complex learning required by modern society. Chapter 2 relates complex learning to the four blueprint components: learning tasks, supportive information, procedural information, and part-task practice. Chapter 3 describes the use of the Ten Steps for developing detailed training blueprints. Then, the Ten Steps are discussed in detail in Chapters 4 through 13. Chapter 14 discusses the teaching/training of domain-general skills in programs based on the Ten Steps, and Chapter 15 discusses the design of summative assessment programs that are fully aligned with the Ten Steps. Finally, Chapter 16 positions the Ten Steps in the field of the learning sciences and discusses future directions.

Acknowledgments

This is the fourth revised edition of this book. To begin, we would like to thank all the teachers, trainers, students, instructional designers, and educators who have bought and used this book to the extent that the publisher asked us to bring out a new, highly revised edition.

Our debt to colleagues and graduate students who in some way contributed to the development of the Ten Steps described in this book is enormous. Without them, this book would not exist. We thank John Sweller, Paul Ayres, Paul Chandler, and Slava Kalyuga for working together on the development of Cognitive Load Theory, which heavily affected the formulation of the Ten Steps. We thank Iwan Wopereis, who helps manage the www.4cid.org website and organize the recurring Dutch 4C/ID User's Day. We thank Ameike Janssen-Noordman and Bert Hoogveld for contributing to the Ten Steps when writing the Dutch books *Innovatief Onderwijs Ontwerpen* [Innovative Educational Design, 2002/2009/2017] and *Innovatief Onderwijs Ontwerpen in de Praktijk* [Innovative Educational Design in Practice, 2011]. We thank our research groups at the Open University of the Netherlands, the Thomas More University of Applied Sciences in Antwerp, Belgium, and Maastricht University. We especially thank our master's and PhD students for the ongoing academic debate and for giving us many inspiring ideas. We thank the many educational institutes and organizations that use the Ten Steps and provide us with useful feedback. Last but not least, we thank all the learners who, in some way, participated in the research and development projects that offered the basis for writing this book.

Finally, we thank Roel Willems, professional content creator and audiovisual teacher. Without his expert help and guidance, it would have been impossible to accurately describe an example 4C/ID blueprint for training a video content producer.

Jeroen J. G. van Merriënboer, Paul A. Kirschner, Jimmy Frèrejean
Maastricht/Hoensbroek, December 2023

Chapter 1

A New Approach to Instruction



When Rembrandt van Rijn painted *The Anatomy Lesson of Dr. Nicolaes Tulp* in 1632, our understanding of human anatomy, physiology, and morphology was extremely limited. The tools of the trade were rudimentary at best and barbarian at worst. Medicine was dominated by the teachings of the church, which regarded the human body as a creation of God, and the ancient Greek view of the four humors (i.e., blood, phlegm, black bile, and yellow bile) prevailed. Sickness was attributed to an imbalance in these humors, and treatments, such as bloodletting and inducing vomiting, aimed to restore this balance. Surgical instruments were basic. A surgeon would perform operations with the most basic tools: a drill, a saw, forceps, and

DOI: 10.4324/9781003322481-1

pliers for removing teeth. If a trained surgeon was not available, it was usually the local barber who performed operations and removed teeth. The trained surgeon was more an 'artist' than a 'scientist.' For example, because there were no anesthetics, surgeons took pride in the speed with which they operated, even amputating a leg in just a few minutes. Progress in anatomy, physiology, morphology, and medical techniques was virtually nonexistent or painfully slow. Although microscopes existed, they lacked the power to reveal bacteria, hindering our understanding of the causes of diseases. This meant that there was also little improvement in medical treatments.

Compare this to today's situation, where hardly a day goes by without new medical discoveries, diseases, drugs and treatments, and medical and surgical techniques. Just a generation or two ago, medicine, medical knowledge and skills, and even the attitudes of medical practitioners toward patients and the way patients approach and think about their doctors were radically different than they are today. It is no longer enough for surgeons to master the tools of the trade during their studies and then apply and perfect them throughout their careers. Competent surgeons today (and tomorrow) must master complex skills and professional competencies—both technical and social—during their studies and never stop learning throughout their careers. This book is about how to design instruction for this complex learning.

1.1 Complex Learning

Complex learning involves integrating knowledge, skills, and attitudes, coordinating qualitatively different constituent skills, and often transferring what is learned in school or training settings to daily life and work. The current interest in complex learning is evident in popular educational approaches that call themselves inquiry, guided discovery, case method, project based, problem based, design based, team based, challenge based, and competency based, many of which have no solid basis in empirical research (Kirschner et al., 2006). Theoretical design models promoting complex learning are, for example, cognitive apprenticeship learning (Collins et al., 1989; Woolley & Jarvis, 2007), first principles of instruction (Merrill, 2020), constructivist learning environments (Jonassen, 1999), learning by doing (Schank, 2010), and the four-component instructional design model (4C/ID; Van Merriënboer, 1997). While these theoretical models differ in many ways, they all emphasize learning tasks based on real-life, authentic tasks as the driving force for teaching, training, and learning. The fundamental idea behind this focus is that such tasks assist learners in integrating knowledge, skills, and attitudes, stimulate the coordination of constituent skills, and facilitate the transfer of what is learned to new problem situations (Merrill, 2020; Van Merriënboer, 2007). Until now, 4C/ID is the only whole-task instructional design model for which a meta-analysis is available (Costa et al., 2021), showing a large positive effect size (Cohen's $d = .79$) on student performance in educational programs developed with it.

The interest in complex learning has grown rapidly since the beginning of the 21st century. It is an inevitable reaction of education and teaching to societal and technological developments and students' and employers' uncompromising views about the value of education and training for updating old knowledge and skills to prevent obsolescence (Hennekam, 2015) and learning new ones. Machines have taken over routine tasks, and the complex cognitive tasks humans must perform are becoming increasingly complex and important (Frey & Osborne, 2017; Kester & Kirschner, 2012). Moreover, the nature of available jobs is changing, necessitating the acquiring and applying new and different skills and quickly making the information relevant to carrying out those jobs obsolete (Thijssen & Walter, 2006). This poses higher demands on the workforce, with employers stressing the importance of problem solving, reasoning, decision making, and creativity to ensure that employees can and will flexibly adjust to rapid changes in their environment.

Two examples might drive this home. Many aspects of the job of an air traffic controller have been technologically automated. But even though this is the case, the complexity of the controller's responsibilities has grown substantially due to the enormous increase in air traffic, the growing number of safety regulations, and the advances in the technical aids themselves (see Figure 1.1). The same is true for family doctors (General Practitioners/GPs),

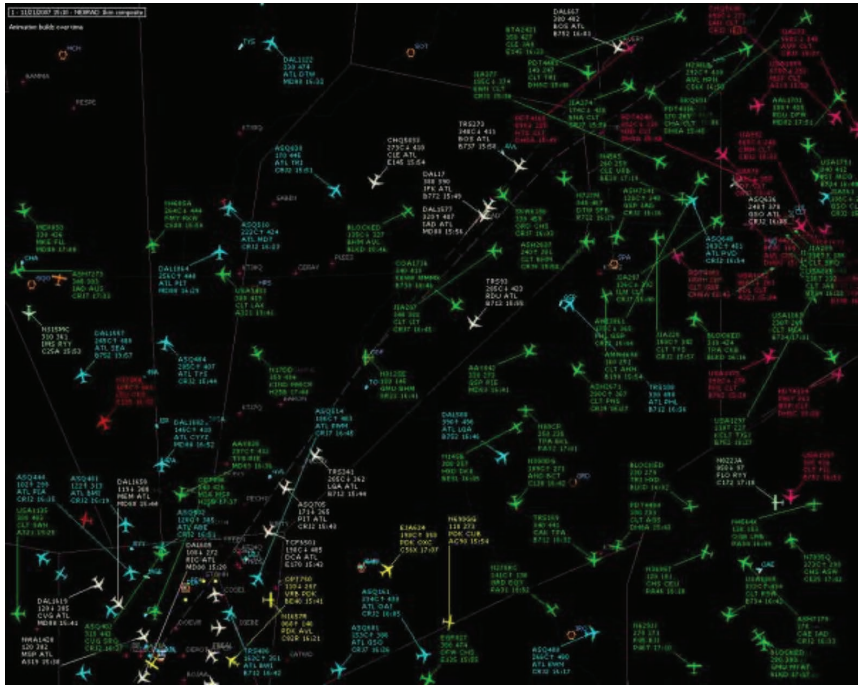


Figure 1.1 Air traffic control screen.

who need to address the physical, psychological, and social aspects of their patients but also encounter a much more diverse clientele with different cultural backgrounds, a flood of new medicines, tools and treatments, and issues dealing with registration, liability, insurance, and more.

The field of education and training has increasingly recognized these new demands posed by society, business, and industry. In response to these demands, there has been a concurrent increase in the attempts to better prepare graduates for the labor market and help them develop *adaptive expertise* (Carbonell et al., 2014). The educational approaches mentioned earlier, emphasizing complex learning and the development of professional competencies throughout the curriculum, strive to reach this goal. However, educational institutes lack proven design approaches. This often results in implementing innovations that undeniably aim at better preparation of students for the labor market but that do so with varying degrees of success (Dolmans et al., 2013).

An often-heard student complaint is that they experience their curriculum as a disconnected set of courses or modules, with only implicit relationships between the courses and an unclear relevance of what they should learn for their future professions and why. This is often complicated by 'flexible' curricula that offer a broad range of possibilities but leave it to the student to choose what and when they want to study without giving them any support or guidance. Often, as a compromise with those instructors who want to 'teach their subject areas,' curricula implement a separate stream in which problems, projects, cases, or other learning tasks are used for the development of complex skills or competencies, hopefully in a—for the student—recognizable and relevant situation. However, even in those curricula, students struggle to link what they are required to do to both the theoretical coursework, which is typically divided into traditional subjects, and what they perceive to be important for their future professions, which often lies at the basis of the learning tasks. Not surprisingly, students have difficulties combining everything they learn into an integrated knowledge-base and employing it to perform real-life tasks and solve practical work-related problems once they have graduated. The whole is inevitably greater than the sum of its parts. In other words, they do not achieve the envisioned and required 'transfer of learning' or 'transfer of training' (Blume et al., 2010).

The fundamental problem facing the field of instructional design is the inability of education and training to achieve this necessary transfer. Design theory must support the development of training programs for learners who need to learn and transfer professional competencies or complex skills acquired in their study to an increasingly varied set of real-world contexts and settings. The *Ten Steps to Complex Learning* (from this point on, referred to as the *Ten Steps*) claims that a holistic approach to instructional design is

necessary to reach this goal (cf. Tracey & Boling, 2013). In the next section, we discuss this holistic design and why it is thought to help improve transfer of learning. Subsequently, we position the instructional design model discussed in this book within the field of learning and instruction, describing its four core components and the ten steps. Finally, we provide an overview of the book's structure and contents.

1.2 A Holistic Design Approach

A *holistic* design approach is the opposite of an *atomistic* one, which continually reduces complex contents and tasks to simpler or smaller elements, such as facts and simple skills. This reduction usually continues to a level where each element can be easily transmitted to the learners through presentation (of facts) and/or practice (of skills). Though such an approach may work well if there are not many interactions between the elements, it does not work well if the elements are closely interrelated. When this is the case, *the whole is more than the sum of its parts* because it contains the elements and the relationships between them. This is the basis of the holistic approach. Holistic design approaches attempt to deal with complexity without losing sight of the interrelationships between the elements taught (Van Merriënboer & Kester, 2008). Using a holistic design approach can offer a solution for three persistent problems in education: compartmentalization, fragmentation, and the transfer paradox.

Compartmentalization

Instructional design models usually focus on one particular learning domain, such as the cognitive, affective, or psychomotor domains. A further distinction, for example, in the cognitive domain, is the differentiation between models for declarative learning, emphasizing instructional methods helping learners to construct conceptual knowledge and models for procedural learning, emphasizing instructional methods helping learners to acquire skills. Everything in a neat compartment. This *compartmentalization*—separating a whole into distinct parts or categories—has had disastrous effects on learning.

Suppose you have to undergo surgery. Would you prefer a surgeon with great technical skills but no knowledge of the human body? Or would you prefer a surgeon with great knowledge of the human body but with two left hands? Or would you want a surgeon with great technical skills but who has a horrible bedside manner and a hostile attitude toward patients? Or, finally, a surgeon with all of the knowledge, skills, and attitudes learned 35 years ago but has not kept them up-to-date? Of course, your answer would be "None of the above". You want a surgeon with up-to-date knowledge and skills who

knows how your body functions (i.e., its anatomy and physiology), is technically dexterous, and has a good bedside manner. This indicates that it makes little sense to distinguish learning domains for professional competencies. Many complex surgical skills simply cannot be performed without in-depth knowledge of the structure and workings of the human body because this allows for the necessary flexibility in behavior. Many skills cannot be performed acceptably if the performer does not exhibit particular attitudes. And so forth. Therefore, holistic design models for complex learning aim to integrate declarative learning, procedural learning (including perceptual and psychomotor skills), and affective learning (including the predisposition to keep all of these aspects up-to-date, including patient skills). So, they facilitate the development of an integrated knowledge base that increases the chance that transfer of learning occurs (Janssen-Noordman et al., 2006).

Fragmentation

Traditional instructional design models use *fragmentation*—the act or process of breaking something into small, incomplete, or isolated parts—as their basic technique (Frèrejean et al., 2022; Van Merriënboer & Dolmans, 2015). Typical of 20th-century instructional design models is that they first analyze a chosen learning domain, then divide it into distinct performance and/or learning objectives (e.g., remembering a fact, applying a procedure, understanding a concept, and so forth), after which different instructional methods are selected for reaching each of the separate objectives (e.g., rote learning, skills labs, problem solving, respectively). In the training blueprint or lesson plan for that domain, the objectives are dealt with one at a time. For complex skills, each objective corresponds with one subskill or constituent skill, and the sequencing of the objectives naturally results in a part-task sequence. Thus, the learner is taught only one or a very limited number of constituent skills at the same time. New constituent skills are gradually added, often each with a different instructional approach, which, in itself, can cause further fragmentation, and it is not until the end of the instruction—if at all—that the learner can practice the whole complex skill.

In the 1960s, Briggs and Naylor (1962; Naylor & Briggs, 1963) reported that this approach is only suitable if little coordination of constituent skills is required and each of the separate constituent skills is difficult for the learners to acquire. The problem with this fragmented approach is that most complex skills or professional competencies are characterized by numerous interactions between the different aspects of task performance, with very high demands on their coordination. In the past half-century, overwhelming evidence has been obtained showing that breaking a complex domain or task down into a set of distinct elements or objectives, then teaching or training each of those objectives without taking their interactions and required

coordination into account, does not work because learners ultimately are not capable of integrating and coordinating the separate elements in transfer situations (e.g., Gagné & Merrill, 1990; Lim et al., 2009; Rosenberg-Kima et al., 2022; Spector & Anderson, 2000). To facilitate transfer of learning, holistic design models focus on reaching highly integrated sets of objectives and, especially, the coordinated attainment of those objectives in real-life task performance.

The Transfer Paradox

In addition to compartmentalization and fragmentation, using a nonintegrated list of specific learning objectives as the basis for instructional design has a third undesired effect. Logically, the designer will select instructional methods that minimize the number of practice items required to learn or master something, the time-on-task spent doing this, and the learners' investment of effort made to reach those objectives; that is, they strive for efficiency. Designing and producing practice items costs time and money, which are often scarce. In addition, the learner does not have unlimited time or motivation to study (as are almost all of us, the learner is a *homo economicus*—one who strives to minimize costs and optimize profits). Take the situation that learners must learn to diagnose three different types of errors (e1, e2, e3) in a complex technical system, such as a chemical factory. If a minimum of three practice items is required to learn to diagnose each error, one may first ask the learners to diagnose error 1, then error 2, and finally, error 3. This leads to the following training blueprint:

e1, e1, e1, e2, e2, e2, e3, e3, e3

Although this 'blocked' practice schedule will be most efficient for reaching the three objectives, minimizing the required time-on-task and learners' investment of effort, it also yields low transfer of learning. This is because the chosen instructional method invites learners to construct highly specific knowledge for diagnosing each distinct error, allowing them to perform in the way specified in the objectives but not to performances beyond the given objectives. If a designer aims at transfer and the objective is that learners can correctly diagnose as many errors as possible in a technical system, then it is far better to train them to diagnose the three errors in a random order. This leads, for example, to the following training blueprint:

e3, e2, e2, e1, e3, e3, e1, e2, e1

This 'random' practice schedule (also called 'interleaving'; Birnbaum et al., 2013) is less efficient than the former one for reaching the three isolated

objectives because it may increase the necessary time-on-task or the investment of effort by the learners. It might even require four instead of three practice items to reach the same level of performance for each separate objective. But in the long run, it yields a much higher transfer of learning! The reason for this increase in transfer is that a random schedule invites learners to compare and contrast the different errors with each other and thus construct knowledge that is general and abstract rather than entirely bound to the three concrete, specific errors. Variability of practice/interleaving requires learners to see the conceptual similarities in problems or tasks that look very different at the surface level and see the conceptual differences between tasks and problems that look very similar at the surface level. This allows learners to diagnose new, not earlier encountered, errors better. This phenomenon—where the methods that work the best for reaching isolated, specific objectives are often not the methods that work best for reaching integrated objectives and increasing transfer of learning—is known as the *transfer paradox* (Helsdingen et al., 2011a, 2011b; Van Merriënboer et al., 1997). A holistic design approach considers the transfer paradox and is always directed toward more general objectives beyond a limited list of highly specific objectives. The differentiation between different types of learning processes should ensure that learners who are confronted with new problems not only have specific knowledge to perform the familiar aspects of those problems but, above all, have the necessary general and abstract knowledge to deal with the unfamiliar aspects of those problems.

To recapitulate, traditional design models usually follow an atomistic approach and, as a result of this, are not very successful in preventing compartmentalization and fragmentation or dealing with the transfer paradox. In contrast, a holistic approach where the tasks are interleaved offers alternative ways of dealing with complexity. Most holistic approaches introduce some notion of *modeling* to attack this problem. A powerful two-step approach to modeling first develops simple-to-complex models of reality or real-life tasks and then 'models these models' from a pedagogical perspective to ensure that they are presented so learners can learn from them (Achtenhagen, 2001). In this view, instruction should begin with a simplified but 'whole' model of reality. This is the essence of what Reigeluth (1992) calls an *epitome* in his elaboration theory: "a very simple kind of case . . . that is as representative as possible of the task as a whole". This is then conveyed to the learners according to sound instructional principles. The Ten Steps offers a broad range of instructional methods to deal with complexity without losing sight of whole, real-life tasks.

1.3 Four Components and Ten Steps

The Ten Steps is a practical, modified, and—as strange as it may sound—simplified version of the 4C/ID model (four-component instructional design;

Miranda et al., 2020; Van Merriënboer, 1997; Van Merriënboer et al., 1992, 2002). Previous descriptions of this model had an analytic-descriptive nature, emphasizing the cognitive-psychological basis of the model and the relationships between design components and learning processes. The Ten Steps, in contrast, is mainly prescriptive and aims to provide a version of the model that is practicable for teachers, domain experts involved in training design, and instructional designers. The focus of this book is on design rather than on learning processes. But for interested readers, some of the chapters include text boxes in which the psychological foundations for particular design principles are briefly explained.

The basic assumption that forms the basis of both 4C/ID and the Ten Steps is that blueprints of educational programs for complex learning can always be described by four basic components; namely, (a) learning tasks, (b) supportive information, (c) procedural information, and (d) part-task practice (see the left-hand column of Table 1.1).

Table 1.1 Four blueprint components of 4C/ID and the Ten Steps.

Blueprint components of 4C/ID	*Ten steps to complex learning*
Learning Tasks	1. Design Learning Tasks
	2. Design Performance Assessments
	3. Sequence Learning Tasks
Supportive Information	4. Design Supportive Information
	5. Analyze Cognitive Strategies
	6. Analyze Mental Models
Procedural Information	7. Design Procedural Information
	8. Analyze Cognitive Rules
	9. Analyze Prerequisite Information
Part-task Practice	10. Design Part-task Practice

The term 'learning task' is used here in a very generic sense: A learning task may refer to a case study that the learners must study, a project that must be carried out, a problem that must be solved, a professional task that must be performed, and so forth. The supportive information helps learners perform nonroutine aspects of learning tasks that often involve problem solving, decision making, and reasoning (e.g., information about the teeth, mouth, cheeks, tongue, and jaw helps a student in dentistry with clinical reasoning; Postma & White, 2015, 2016). The procedural information enables learners to perform the routine aspects of learning tasks; that is, those aspects of the learning task that are always performed in the same way (e.g., how-to instructions for measuring blood pressure help a medical student with conducting physical examinations). Finally, part-task practice pertains to additional practice; that is, *overlearning* of routine aspects to help learners develop a high level of automaticity of these aspects and improve

their whole-task performance (e.g., practicing cardiopulmonary resuscitation [CPR] allows a nurse or 'first responder' to be better prepared for emergencies).

As indicated in the right-hand column of Table 1.1, the four blueprint components directly correspond with four design steps: the design of learning tasks (Step 1), the design of supportive information (Step 4), the design of procedural information (Step 7), and the design of part-task practice (Step 10). The other six steps are auxiliary to these design steps and are only performed when necessary. In Step 2, performance assessments are designed based on objectives and standards for acceptable performance. Step 3 organizes learning tasks in simple-to-complex levels, ensuring that learners initially work on simple tasks (i.e., not complex) and then smoothly increase in complexity. Steps 5 and 6 may be necessary for in-depth analysis of the supportive information that would help learn to carry out the nonroutine aspects of learning tasks. Steps 8 and 9 may be necessary for in-depth analysis of the procedural information needed for performing routine aspects of learning tasks.

It should be noted that real-life design projects are never a straightforward progression from Step 1 to Step 10. New results and decisions will often require the designer to reconsider previous steps, causing iterations in the design process. One may design a few learning tasks, in a process of rapid prototyping, before designing the complete educational program. In addition, particular steps may be excessive for particular design projects. As a result, zigzagging (see Section 3.2) between the Ten Steps is common. Then, the trick of the trade is to keep a good overview of all—intermediate—design and analysis products and their relations to the ultimate training blueprint. Computer-based tools will be beneficial to carry out larger design projects, as they facilitate the systematic development of an educational blueprint and help designers to keep the required overview of the whole project—even when they zigzag between different steps (Van Merriënboer & Martens, 2002).

1.4 Using the Ten Steps

The Ten Steps can be seen as a model of instructional design specifically directed toward programs of vocational and professional education (both at the secondary and higher education level), profession-oriented university programs (e.g., medicine, business administration, law), and competency-based training programs in business, industry, government, and military organizations. In higher education, it is used to design courses for research and other complex skills in various domains (Bastiaens et al., 2017; Frèrejean et al., 2019). Yet, applications of the model also appear in general secondary education and even primary education for teaching complex skills in traditional school subjects (e.g., Melo, 2018; Melo & Miranda, 2015; Wade

et al., 2023) and specific skills (e.g., Costa & Miranda, 2019; Güney, 2019a, 2019b; Linden et al., 2013; Maddens et al., 2020; Zhou et al., 2022). Finally, it is used in continuous professional development; for example, in the fields of teacher training (Frèrejean et al., 2021; Kukharuk et al., 2023; Meutstege et al., 2023) and medical training (Kolcu et al., 2020). The model will typically be used to develop training programs of substantial duration—ranging from several weeks to several years. In terms of curriculum design, the model will typically be used to design a—substantial—part of a curriculum to develop one or more professional competencies or complex skills.

While the Ten Steps is strongly informed by decades of research on learning and cognitive psychology and offers many clear recommendations for design activities, it is essential to consider the unique needs of each situation when selecting an instructional design model. When deciding to use the Ten Steps, the following considerations may help:

- We recommend the Ten Steps when your instructional objective is to develop one or more complex skills requiring integrating knowledge, skills, and attitudes and coordinating constituent skills. Such skills involve substantial amounts of problem solving, reasoning, or decision making. If your instructional objective is to develop only knowledge, a single skill, or an attitude, the integration and coordination central to the Ten Steps are less relevant.
- We recommend the Ten Steps if your goal is to achieve transfer of learning and prepare learners for real-life situations or workplace activities, especially in fields characterized by rapid changes or frequent occurrences of unfamiliar tasks. In contrast, the Ten Steps approach can be excessive for tasks involving little problem solving, reasoning, and/or decision making; for example, for teaching procedural tasks such as stocking shelves or calibrating a machine. Alternatively, if the course does not aim for clear, real-world applications, the Ten Steps might not offer significant advantages over more traditional methods or may even be disadvantageous.
- We recommend the Ten Steps if your learning environment allows learners to work on whole tasks in your domain, such as group or project work, simulations, computer-based tasks, etc. Applying the Ten Steps becomes challenging if this is not possible; for example, if your building has large lecture halls but no small group rooms or adequate computer facilities for working on learning tasks.
- We recommend the Ten Steps if your program can have sufficient length to develop the intended complex skill or skills. A rule-of-thumb is that complex skill development in programs shorter than 100 hours is unlikely. While you could apply the Ten Steps to design a short program, learners are unlikely to achieve mastery and transfer of learning when given insufficient time to practice and reflect.

- We recommend the Ten Steps if you have sufficient resources for development. Designing the four components may require the involvement of educational specialists, subject-matter experts, practitioners, teachers, students, and multimedia developers (see Chapter 16). While rapid instructional design approaches exist and can be helpful, this book will make apparent that many design decisions require a robust analysis of the situation and an understanding of which methods are effective in that situation. Working fast or with limited information to inform your decisions can bring the risk of sacrificing effectiveness.
- Finally, we recommend the Ten Steps if you have sufficient autonomy to make design changes in your current program or decisions about a new program. Without control over course length, assessment methods and moments, or structure and planning, your opportunities for adequately applying the design principles are constrained. Another challenge may occur in subject-based educational programs with a strong separation of teams or persons responsible for each subject, where it may require effort to integrate subject domains into whole tasks.

The Ten Steps is a versatile instructional design model applicable across various educational contexts. It is important to recognize that the Ten Steps is not a rigid recipe but a systematic approach that allows creativity and flexibility. No two applications of the Ten Steps are identical, and designers frequently make changes to accommodate the constraints of their situation. Such adaptations are not only common but also expected. The previous considerations serve as a guide, signaling situations that might require more extensive deviations.

The remainder of this book describes the Ten Steps in 16 chapters. Chapters 2 and 3 introduce the four blueprint components and ten steps. Chapters 4 to 13 make up the main part of this book. Each chapter contains a detailed explanation of one of the ten steps. Chapters 14 and 15 describe, in order, how domain-general skills can be taught in educational programs designed with the Ten Steps and how programmatic assessment can be applied in those programs. Chapter 16 discusses future directions in the field of complex learning.

Practitioners in the field of instructional design may use this book as a reference guide to support their design of courses, materials, and/or environments for complex learning. To make optimal use of the book, it may be helpful to consider the following points:

- It is probably best for all readers to study Chapters 1, 2, and 3 first, regardless of their reason for using this book. They introduce the four blueprint components and the Ten Steps.

- Chapters 4 through 13 describe the Ten Steps in detail. You should always start your design project with Step 1, but you only need to consult the other chapters if these steps are required for your project. Each chapter starts with general guidelines that may help you decide whether the step is relevant to your project.
- The next two chapters are relevant for readers with a specific interest in training domain-general skills (Chapter 14) or designing programs of assessment (Chapter 15). Minimum requirements for assessment (performance assessment of learning tasks; Step 2 in Chapter 5) and teaching domain-general skills (task selection by the learner in on-demand education; Step 3 in Chapter 6) are already an integral part of the Ten Steps.

If you are a student in the field of instructional design and want to broaden your knowledge of the design of training programs for complex learning, we advise you to study all chapters in the order in which they appear. For all readers, whether practitioner or student, we tried to make the book as useful as possible by including the following:

- Each chapter ends with a *Summary* of its main points and design guidelines.
- Key concepts are listed at the end of each chapter and included in a *Glossary*. This glossary contains pure definitions of terms that might not be familiar and, in certain cases, may be more extensive (in the case of seminal or foundational concepts, theories, or models) and contain background information. In this way, the glossary can help you organize the main ideas discussed in this book.
- In several chapters, you will find *Boxes* in which the psychological foundations for particular design decisions are briefly explained.
- Two *Appendices* with example materials are included at the end of this book.

Four Blueprint Components

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Step 1

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